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NPS/ 001 / 014 – Technical Specification for Overhead Line Fault Passage Indicators

1. Purpose

This document details the requirements for pole mounted remote FPI's (Fault Passage Indicators) for use on the CE Electric UK HV distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NPS/001/013	1.0	Technical Specification for Overhead Line Fault Passage Indicators
DSS/007/023	1.0	Specification for OHL Fault Passage Indicators
NEDL/MS – O/40	2004	Technical Specification for Overhead Line Fault Passage Indicators

2. Scope

This document applies to equipment that identifies the passage of fault current in 3-phase 3-wire and single-phase 2-wire overhead line systems of earthed and unearthed construction operating between 6.6kV and 33kV. 2 types of unit are covered, those with the ability to identify the fault status locally via a visual flashing light and those which in addition are able to simultaneously send a signal to a centralised remote monitoring centre via an inbuilt GSM/GPRS communications system.

Control or field engineers shall be able to simply follow the "flashing" lights or GSM/GPRS messages along the feeder. The fault will lie in the section beyond the last instrument to flash and before the next unit not to flash.

With the exception of battery replacement every 10 years, this item shall be designed to be maintenance free for an operational life expectancy of 40 years.

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3. Technical Requirements

3.1 General Requirements

3.1.1 Description

The FPI shall be designed to continuously monitor the residual current and voltage fields from the overhead line conductors. The FPI shall be triggered when fault current passes in the line conductors. In the "alarmed" state it shall flash a high intensity light or red LED (dependant upon fault type).

3.1.2 Technical Particulars

System Operating Voltage (kV)	6.6 – 33kV
11kV Sensitive Earth Fault Level (A)	15A
20kV Sensitive Earth Fault Level (A)	30A
Live/Dead Line Trip Level (A)	7 – 20A
Temperature Range (°C)	-40°C to +70°C
Humidity Range	0 – 100%
IP Rating	IP67
High intensity flash Rate (s)	10- 12 seconds
High intensity light brightness (Candela)	3000 Mill Candela
Voltage Reset Time (s)	60 seconds
Battery Type	Lithium
Minimum battery life under normal operating conditions	10 years
Battery Shelf Life	15 years
Installation Distance from Conductors (meters)	3m
Unit Impact Resistance (Joules)	2 Joules

Note: - Appendix 2 shall be completed for each unit offered

3.2 General Conditions

3.2.1 System Conditions

The equipment shall be suitable for use on distribution systems in which the neutral point is earthed either directly with typical maximum earth fault passage levels of 2000A for 1.5 secs or by means of a neutral earth resistor in which the earth fault passage levels have been limited to 1200A for 1.5 secs. On certain parts of the distribution system, the above earthing methods have been replaced by or supplemented with arc suppression coils.

The arc suppression coils have been installed to manage earth faults that are transient in nature. The protection system has been designed in such a way that the coil remains in circuit for a specific period of time limiting the flow of fault current. This action negates the requirement for an immediate circuit breaker operation and thus allows the system to operate normally for transient faults. If the fault is sustained, the protection system reverts back to the more traditional resistance earthed protection environment detailed above.

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Typically the arc suppression coils have been designed to be in circuit for the following times before reverting to resistance earthing :

11kV - up to 30secs.

20kV - up to 5 secs.

A review carried out on the CE Electric UK distribution system has identified that under certain fault conditions on primary substations fitted with arc suppression coils, a fault on one circuit may lead to a displaced neutral condition and as a result the production of unbalanced capacitive currents flowing down non faulted circuits. Since this can result in false operation of the FPI's, the FPI's equipped with GSM/GPRS communications shall be capable of having their earth sensitivity levels remotely updated during the next heartbeat health check.

3.2.2 Environmental Conditions

Fault passage indicators shall be completely weatherproof and shall operate under conditions of shock and vibration normally encountered in service. Indicators shall be environmentally tested to IP67.

They shall be suitable for continued reliable operation in the temperature range -40°C to +70°C with humidity levels of between 0 - 100%. The equipment enclosure shall be UV stable.

Variation in system frequency of 2Hz from the nominal frequency of 50Hz shall not cause a variation of more than 5 per cent in the operation or resetting characteristics of a fault passage indicator.

3.2.3 Temperature

The unit shall demonstrate stability of operation for the equipment on both Earth Fault and Instantaneous threshold trip levels against the stated temperature range.

3.2.4 Vibration Endurance Testing

The equipment shall be suitably designed to withstand all likely vibrations present on a wood pole supporting structure.

3.2.5 UV Stability

The equipment shall be manufactured from Ultra Violet stable materials including all manufactures markings as detailed in clause 3.7.6

3.3 Modes of Operation

3.3.1 Desired Operating Modes

The unit shall be suitable for recording permanent, intermittent and transient faults. Since many overhead line faults are the nuisance intermittent or the transient type both of which cause auto- reclose operations and no obvious signs of damage, the FPI shall be designed to interpret the difference between these fault types and respond accordingly.

A passage of fault current combined with a sustained loss of voltage shall result in a high intensity light flashing for a designated period of time (as detailed in section 3.3.2 and 3.3.3) depending upon the fault

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type. This time period will provide sufficient time for field staff to inspect the indicators on the system and patrol the faulty section of network. A passage of fault current without a sustained loss of voltage shall not result in the flashing of the high intensity light.

3.3.2 Permanent Fault

If the fault is permanent (indicated by a sustained loss of voltage) then any upstream FPI's shall be designed to alarm with the flashing of the high intensity light until the voltage is restored or for a period of 3 hours, whichever occurs first. To save battery capacity, once the high intensity light stops flashing the red LED or low energy identifier will commence flashing and continue for 24 hours. At the time of the fault, the GSM/GPRS module (if fitted) shall signal back to the basestation to record the permanent fault event. An additional signal shall also be generated when the units are reset and normal system voltage is restored.

3.3.3 Intermittent and Transient Faults

In the instances where there has been a successful auto-reclose and the system volts have been restored, only the red LED will flash for a period of 24 hours. The high intensity light shall have been reset by the system voltage after 60 seconds.

At the time of the fault event, the GSM/GPRS module (if fitted) shall record the transient fault event. There should be a user-settable option as to whether such events are immediately communicated to the base station or stored and transmitted to the base station during the next heart-beat contact.

3.4 Earth Fault Sensitivity Levels

In general the FPI's shall be designed to be graded with the protection curves of adjacent ground or pole mounted reclosers, thus operating effectively at all fault levels down to as low as 7Amps.

Initially the FPI's shall be supplied preset with two different earth fault sensitivity levels: -

- 11kV units shall be set with default earth fault sensitivity levels of 15A
- 20kV units shall be set with default earth fault sensitivity levels of 30A

The FPI's shall be designed such that earth fault sensitivity level can be amended from it's default value to a user entered value. If fitted with GSM/GPRS communications, this amendment shall be achievable via the remote signaling system. Typically during the heartbeat health check, the basestation shall be able to re-programme this setting if required.

FPI's shall be designed to discriminate between magnetic inrush current and fault current.

3.5 Battery Life

FPI's shall utilise a single monitored battery suitably sized to provide the specified operation for a minimum of 10 years, assuming the following typical operating conditions:

- Average ambient temperature 16°C giving average self discharge of 2% remaining capacity per annum.
- 2 Weekly health check GSM messages..
- Minimum shelflife of 2 years before installation on the distribution system.

FPI's shall incorporate a battery test facility allowing field staff to confirm the battery health status on site.

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3.5.1 Battery Disposal

Manufacturers shall offer CE Electric UK a facility for the disposal of recovered batteries when batteries reach their end of life.

3.6 Communications

3.6.1 Communications Medium

Note

CE Electric UK utilises the I-Host Data Management Software to monitor and manage the remotley located FPI's that have been specified to include a GSM/GPRS communications interface. A further interface has been developed to allow this information to be interfaced with the company's GE Harris Enmac NMS control system.

Where specified, the FPI's shall incorporate a communications modem that is capable of communicating on both the UK GSM System and the GPRS system.

The outstation FPI unit shall be designed to communicate via the GSM/GPRS network system into the above I-Host Data Management Software to report that there has been a passage of fault current on the overhead line conductors at that location on the distribution network.

3.6.2 Functionality

- The modem unit shall be designed to accept a SIM card that operates with any network provider.
- The system shall be designed to allow fault alarms to be sent via SMS Text messages, or Data Calls. By default the SIM cards will be programmed for GPRS DATA calls.
- The modem unit shall be capable of generating multiple alarm events eg permanent faults, transients faults, system reset after permanent fault, signal strength, battery condition, and next heartbeat time/date.
- Once a fault has been detected, the GSM/GPRS modem with the FPI shall send an alarm to the basestation PC (Software shall provide to interface a receiving modems with a standalone PC). The software shall be designed to populate the customer's chosen database.
- The FPI shall be capable of receiving updated earth fault settings via the communication system.
- The FPI unit shall be capable of generating an unsolicited heartbeat message. The scheduled timing of the message being user definable between day/week/month. At the time for the scheduled heartbeat check there shall be a maximum of 3 attempts with 2 minute intervals for a successful transmission. If the heartbeat message is not received due to communication failure etc. a PC type base station will alarm 15 minutes after first heartbeat due time.
- The FPI shall be able to time stamp events with reference to the Server's clock at time message received by Server and to give differential event stamping to within 1 sec.
- The units shall be designed to have no visable antenna or aerial.

3.6.3 Modem

The modem shall comply with the following standards: -

ETSI 300342 – 2
ETSI 31489 Part 7

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3.6.4 Software Requirements

See clause 3.6.1 for details of existing FPI management software being utilised by CE Electric UK.

3.6.5 On Board Diagnostics

FPI's equipped with GSM/GPRS communications shall incorporate on board diagnostics that carries out a scheduled low battery health check. The unit shall be programable to allow an alarm event to be generated when the remaining battery life reaches a user definable limit.

3.7 Testing

3.7.1 Electromagnetic Compatability

Fault passage indicators shall have been fully EMC tested to the following standards:

BS EN 61000-6-3 :2007 : Emission standards for residential, commercial and light-industrial environments
BS EN 61000-6-2 :2005 : Immunity standards for industrial environments.
BS EN 60801-2 :1993 : Electrostatic discharge requirements
(Min Values Air 4kV, Contacts 8kV)

BS EN 61000-4-3 :2002 : Radiated, radio-frequency, electromagnetic field immunity test
(Min Values 27Mhz - 1Ghz @ 10V/M)

IEC 61000-4-4 : 2004 : Fast Transients/Bursts
(Min Values +/- 1kV)

3.7.2 Communications - GSM / GPRS Electomagnetic Compatability Tests for the communication equipment

The modem unit shall be compatible with the following generic EMC standards:

BS EN 61000-6-3: 2007 - Generic Emmisions standard
BS EN 61000-6-2: 2005 - Generic Immunity standard
BS EN 61000-4-2: 1995 - Electrostatic discharge immunity Test
(Requirement Air 8kV, Contact 4kV)
BS EN 61000-4-3: 2006 Radiated radio Frequency, Electromagnetic Field Immunity. (Requirement 80MHz - 1GHz @ 10V/m)
BS EN 61000-4-6: 2007 Immunity to conducted disturbances induced by radio frequency fields
(Requirement 150kHz - 80MHz @ 10V/m)
BS EN 61000-4-4: 2004 Electrical Fast Transient/Bursts Immunity
(Requirements +/- 1kV).

3.8 General

3.8.1 Equipment Location

The equipment shall be designed for permanent attachment to wood poles with mounting locations approx 3.0m symetrically below the line conductors. They shall operate on flat, triangular and wishbone formation lines operating in three or single phase modes of operation. There shall be no physical connection between the FPI and the line or conductors.

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3.8.2 Pole Fixing

The CE Electric UK preference for fixing the FPI's to the pole shall be by means of a two point fixing bracket complete with keyhole style fixing holes suitable for 75mm long M10 or M12 coach screws.

3.8.3 Product Markings

All units shall be marked with the following information:-

External Markings

- Name of manufacturer
- Year of manufacture
- Serial Number

Internal Markings

- Serial Number
- Manufacturing Batch Code
- Battery Installation Date Label (To allow information update when battery is changed)

Additional markings on Non Communicating FPI's

Non-communicating FPI's shall be marked with the voltage, programmed default Earth sensitivity settings and the unit serial number with a date of manufacture.

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4. References

4.1 External Documentation

Reference	Title
BS EN 61000-6-3:2007	Generic standards - Emission standards for residential, commercial and light-industrial environments
BS EN 61000-6-2 :2005	Generic standards - Immunity standards for industrial environments.
BS EN 60801-2 : 1993	Electromagnetic compatibility for industrial-process measurement and control equipment. Electrostatic discharge requirements
BS EN 61000-4-2 :1995	Electrostatic Discharge Immunity test
BS EN 61000-4-3 :2002	Testing and measurement techniques. Radiated, radio-frequency, electromagnetic field immunity test
BS EN 61000-4-4:2004-07	Testing and measurement techniques - Electrical fast transient/burst immunity test
BS EN 61000	Electromagnetic compatibility

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
FPI	Fault Passage Indicators
NMS	Network Management System
GSM	GSM (Global System for Mobile communication) is a digital mobile telephony system that is widely used in Europe and other parts of the world. GSM digitizes and compresses data, then sends it down a channel with two other streams of user data, each in its own time slot. It operates at either the 900 MHz or 1800 MHz frequency band.
GPRS	General Packet Radio Services (GPRS) is a packet-based wireless communication service that provides data rates from 56 up to 114 Kbps and continuous connection to the Internet for mobile phone and computer users. GPRS is based on Global System for Mobile (GSM) communication system.
Sim Card	GSM phones use a Subscriber Identity Module (SIM) smart card that contains user account information.

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G.Hammel	Overhead Standards Manager	G.Hammel	11/01/08

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M.Nicholson	Head of System Strategy	M.Nicholson	11/01/08

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
J.Morrell	Network Investment Engineer	J.Morrell	11/01/08

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	18/01/08

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards Manager	C.Holdsworth	11/01/08

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	11/01/08

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Appendix 1 – Addendum to Suppliers Requirements

A1 Battery Life

Manufacturers shall provide sample calculations demonstrating the design capacity of the battery over the 10 year service life. The calculations shall include all assumptions in particular the units remaining capacity for dealing with fault events after removing the estimated drain detailed in section 3.4 of this specification.

A2 Protection and Packaging

The Contractor shall ensure that each item is suitably protected and packaged to maintain its "fit for service" status prior to installation on our system.

A3 Packaging Markings

Each package must be marked with the following information:

- (i) Manufacturers name.
- (ii) Manufacturer's product type reference.
- (iii) Description of item.
- (iv) Date of supply.
- (v) Purchasing Companies Stock Cat Number.
- (vi) Weight of each unit or package.

A4 Installation, Operation and Maintenance Instructions

Appropriate installation, operation and maintenance instructions shall be supplied with each unit. They must also detail the commissioning procedures.

Sufficient numbers of any special equipment or tools needed to test and commission the FPIs shall be provided so as not to hamper the efficient testing and installation of the FPIs.

A5 Item Schedule

Item Number	Cat Number	Description
1	Tba	Non Communicating Overhead Line FPI – programmed for 11kV Earth Fault Sensitivity
2	Tba	Non Communicating Overhead Line FPI – programmed for 20kV Earth Fault Sensitivity
3	Tba	Dual Mode GSM/GPRS Overhead line FPI

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Appendix 2 – Schedule of Suppliers Technical Data

Description	Parameters Available on offered equipment
System Operating Voltage (kV)	
11kV Sensitive Earth Fault Level (A)	
20kV Sensitive Earth Fault Level (A)	
Live/Dead Line Trip Level (A)	
Initial Time Delay	
Voltage Reset Time (s)	
Temperature Range (°C)	
Humidity Range	
Enclosure IP Rating	
Enclosure construction material	
Quiescent drain	
High Intensity Light Flash Rate (s)	
High Intensity Light Power Demand – 3 hour flash period (mA.hrs)	
Low current demand flash –	
Transient fault Power demand – 24 hour flash period (mA.hrs)	
Battery Type	
Battery Life	
Battery Shelf Life	
Installation Distance from Conductors (meters)	
Unit Impact Resistance (Joules)	
Typical fault handling capability for – Units with communication set at 2 weekly heartbeat - (number of faults)	

To be completed for each unit being offered

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Appendix 3 – Self Certification Conformance Declaration

Fault Passage Indicators for overhead lines shall comply with the latest issues of the IEC's and British Standards quoted within this specification.

Key elements from the above standards and this specification have been quoted to amplify and/or clarify the requirements of those Standards. This check sheet identifies the particular clauses of the aforementioned Standards relevant to overhead line fault passage indicators.

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes for each conductor.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Note

Separate Self Certification Conformance Declaration sheets shall be completed For each product being offered.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Electromagnetic Compatibility			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
BS EN 61000-6-3:2007	Emission standards for residential, commercial and light industrial environments.		
BS EN 61000-6-2:2005	Immunity Standards – Immunity for industrial environments		
BS EN 60801-2:1993	EMC for industrial-process measurement and control equipment – Electrostatic discharge requirements Min values Air – 4kV Contacts – 8kV		
BS EN 61000-4-3:2002	EMC – Radiated, radio- frequency, electromagnetic field immunity test Min Values 27Mhz – 1Ghz @ 10V/m 80Mhz – 1Ghz @ 10V/m		
IEC 61000-4-4:2004	EMC – Electrical fast transient/burst immunity test Min values +/- 1kV		
BS EN 61000-4-6:2007	EMC - Immunity to conducted disturbances induced by radio frequency fields Min Values 150khz – 80Mhz @ 10V/m		

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Specific requirements within this specification			
Clause/Sub-clause	Requirements	Conformance Code	Remarks
Clause 3.2.2	Environmental conditions Temp -40°C to +70°C Tested to IP67		
Clause 3.2.3	Temperature stability during fault conditions		
Clause 3.3	Desired Operating modes Differentiate between permanent, intermittent and transient faults for installation on main lines and tee's.		
Clause 3.4	Adjustable Earth fault Sensitivity 11kV – 15Amps 20kV – 30Amps		
Clause 3.5	Battery Life 10 years, based on two weekly heartbeat calls		
Clause 3.6.1	Communication requirements Capable of GSM and GPRS A proven capability to interface into the I Host Data management software.		
Clause 3.6.2	Functionality requirements		
Clause 3.8.3	Product markings External Markings Internal markings		